Skills Summary

Reading and Plotting Points on a Grid

Use this reference while completing your worksheet or when studying.

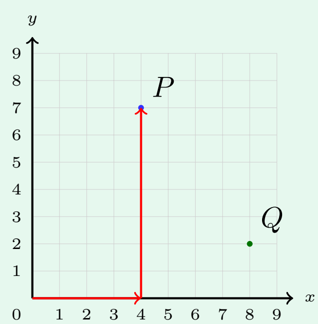
Skill 1 — Reading a point: across first, then up

An ordered pair is written (x, y). The first number says how far to travel **across** from the corner marked 0; the second says how far to travel **up**.

Always in that order. $(3, 7)$ and $(7, 3)$ use the same two numbers and land in completely different places, which is why they are called *ordered* pairs.

A coordinate of 0 is a real instruction: it means *do not travel that way at all*. The corner itself is $(0, 0)$.

Example: Write point P as an ordered pair.



Step 1. A point is read as a journey from the corner, and the journey always goes across before it goes up — so trace the red arrows in that order.
Step 2. Across the bottom to sit under P: four squares. Then straight up to P: seven squares.

$\Rightarrow \quad P = (\mathbf{4}, \mathbf{7})$

Try it: Write the green point Q on the same grid as an ordered pair.

check: $Q = (\mathbf{8}, \mathbf{2})$

Skill 2 — Plotting a point and measuring a line

To plot (x, y): put your finger on 0, slide across x squares, then go up y squares, and mark the spot.

To measure a straight line between two points: if the line is *upright*, both points have the same x, so subtract the two y-coordinates. If the line is *flat*, both points have the same y, so subtract the two x-coordinates. Only one of the two coordinates does any work.

Example: Plot $R(5, 2)$ and $S(5, 7)$. How many squares long is $\overline{RS}$?
Step 1. Both points are five squares across, so the line joining them is upright and only the up-and-down numbers can differ — subtract those.
Step 2. y of S minus y of R: $7 - 2$.
$\Rightarrow$ $\overline{RS} = \mathbf{5}$ squares

Try it: Plot $T(3, 3)$ and $U(3, 9)$. How many squares long is $\overline{TU}$?

check: **6** squares

Skill 3 — Two rules make a list of points

Run two counting rules side by side. The number from the first rule is the x, the number from the second rule is the y, and together they make one ordered pair.

Plot the pairs in order and they fall on a straight line. If one rule adds 1 each time and the other adds 3, then y climbs three times as fast as x — so y is always three times x, and you can jump straight to any pair without writing out the rows in between.

Example: Rule A: start at 0, add 1. Rule B: start at 0, add 3. What is y when $x = 2$?
Step 1. Rule B climbs three steps for every one step of Rule A, so instead of listing rows, multiply the x by three.
Step 2. The pairs run $(0, 0)$, $(1, 3)$, $(2, 6)$, and $3 \times 2 = 6$ agrees.
$\Rightarrow$ $y = \mathbf{6}$, so the pair is $(2, 6)$

Try it: Same two rules. What is y when $x = 4$?

check: $y = \mathbf{12}$, so the pair is $(4, 12)$

(!) Watch out: the commonest mistake on a grid is doing the two journeys in the wrong order — up first, then across. Say the pair out loud as you plot it: “across four, up seven.” If a point lands somewhere unexpected, check the order before checking anything else.